SBMS ML SUBSYSTEM DOCUMENTATION

Prepared: 2026-11-02 **Revision described:** `fix/sbms-completion-work` @ `d6b5032` **Sources read:** `ml-service/feature_engineering.py`, `ml-service/predict.py`, `backend/src/services/duplicateCheck.js`, `backend/src/services/mlDuplicateReview.js`, `backend/src/services/duplicateAggregation.js`, `backend/src/controllers/students.controller.js`, `docs/ml-decision-sheet-2026-10-30.md`, `docs/ml-evaluation-baseline-2026-10.md`, `docs/ml-training-integrity-2026-10-30.md`, `ml-service/reports/sbms_dataset_experiment.md`.

---

1. What is deployed, what is research

Item	Status	Identity
Production classifier	**Deployed and called on every registration**	`XGBClassifier` inside a scikit-learn pipeline, six features, artifact `ml-service/model/xgb_duplicate_model.joblib`, model version `xgboost:6ec9f2a3cbe6ed0dcf80c79b15f47364d9f4f7bd3e27da25c690d0436bb0a6f9`
Random Forest	**Not deployed.** Research candidate only	`ml-service/model/research/rf_attachment_style_candidate.joblib`; the application never loads it
Research ten-feature model on the supplied archive	**Not deployed and structurally incompatible with the serving path**	`ml-service/experiments/sbms_dataset/artifacts/sbms_dataset_xgb_research_only.joblib` (183 445 bytes, separate hash)
Review cut-off	**Deployed, provisional**	0.50 (`ML_REVIEW_SCORE_THRESHOLD`), `thresholds_validated: false`
High-priority cut-off	**Deployed, provisional, presentation only**	0.70 (`DUPLICATE_HIGH_PRIORITY_THRESHOLD`)
Human adjudication	No adjudicated pair exists	`ml-service/evaluation/reviewed_pairs_template.csv` is header-only

The approved proposal specified a Random Forest over four features. What runs is an XGBoost classifier over six. That is a design deviation (G-21), reported as a deviation rather than reconciled, because a reader must not be told the proposal was followed when it was not.

2. Integration architecture

Element	Implementation	Consequence
Where scoring happens	Node service `backend/src/services/duplicateCheck.js` spawns `python predict.py --server` and exchanges newline-delimited JSON over stdio	No HTTP hop, no extra port, one container
Request shape	`{pairs: [{first, second}, ...]}` with raw student records	The feature builder runs in Python, so training and serving share one implementation
Response shape	`{model_version, results: [{score, features}]}`	Score and the six feature values come back together and are stored with the case
Model version	SHA-256 of the artifact file, prefixed `xgboost:`	A stored case names the exact bytes that scored it
Failure mode	Predictor unavailable or timed out → registration aborts with **503** and no student row is created	An unscreened beneficiary cannot be registered silently (NFR-20)
Batching	All candidates for one registration are sent in a single request	One subprocess round trip per registration, not per pair
Training calls in the serving path	None: `predict.py` contains no `.fit(`, `GridSearchCV`, `joblib.dump` or `train_test_split`, and the assertion is enforced by `ml-service/tests/test_training_integrity.py`	Live registrations cannot change the model

The Python process is long-lived for the life of the API process and is restarted with it. It is not a REST service, and it has no authentication of its own, because it is reachable only through the Node process's stdin.

3. Feature specification (the six model inputs)

`ml-service/feature_engineering.py` is the single source of truth; the training scripts and `predict.py` both import it.

#	Feature	Type and range	Definition	Missing-value behaviour
1	`jaro_winkler`	float 0-1	**Maximum** of Jaro-Winkler similarity on the concatenated name tokens and on the alphabetically sorted tokens	0.0 if either side is missing
2	`dob_match`	float in {0, 0.5, 0.8, 1.0}	1.0 same date; 0.8 one day apart; 0.5 within seven days; 0.0 otherwise	0.0 if either side is missing or unparseable
3	`school_match`	int {0, 1}	Exact match after removing non-alphanumerics and case-folding	0
4	`guardian_match`	float 0-1	Identifier similarity after stripping shared prefixes (`guardian`, `guard`, `upi`, `student`, `bc`, `g`); 1.0 when equal	0.0
5	`upi_match`	float 0-1	As above for the UPI	0.0
6	`birth_certificate_match`	float 0-1	As above for the birth certificate number	0.0

Accepted date patterns are `%Y-%m-%d`, `%Y/%m/%d`, `%d/%m/%Y`, `%d-%m-%Y` and `%d.%m.%Y`. Kenyan day-first convention is assumed; an ambiguous US-style date is deliberately not treated as a confident match, which is the behaviour asserted by `backend/tests/predictorDateNormalization.test.js` (6 tests).

"Missing" is a defined set, not merely an empty string: `''`, `?`, `NA`, `N/A`, `NONE`, `NULL`, `UNKNOWN`, `NOT PROVIDED`, `NOT AVAILABLE`, any value that is all zeros with length four or more, and any value with no alphanumeric characters. A shared placeholder such as `NOT PROVIDED` on both rows therefore produces similarity 0, not 1 — the defect class the integrity checker explicitly tests for.

4. Prediction contract

Function	Input	Output	Used by
`predict(data)`	one dict of the six named features	`{probability}` rounded to four decimals	Unit tests, one-off scoring
`predict_many(list)`	list of feature dicts	list of `{probability}`	Batch paths
`predict_pairs(pairs)`	`[{first, second}]` raw records	`[{score, features}]`	The Node service, via the `--server` protocol
`predict_duplicate(first, second, threshold=0.5)`	two raw records	`{duplicate_probability, is_duplicate, features}`	Not called by the API; a convenience wrapper used in the service tests
`model_version()`	—	`xgboost:<sha256 of artifact bytes>`	Every stored case and every registration response

`probability` is the raw positive-class score of the classifier. It is **not** a calibrated probability, and the registration response says so explicitly with `score_is_calibrated_probability: false`. A score of 0.6 does not mean "60% chance of being a duplicate"; it means "above the provisional review cut-off".

5. Thresholds in force

Threshold	Value	Environment variable	Effect	Validation status
Model review cut-off	0.50	`ML_REVIEW_SCORE_THRESHOLD`	A pair at or above this score becomes a review case	`thresholds_validated: false`; F1 is flat from 0.30 to 0.60, so no value in that band is defensible over another on the available evidence

Threshold	Value	Environment variable	Effect	Validation status
High-priority cut-off	0.70	`DUPLICATE_HIGH_PRIORITY_THRESHOLD`	Presentation only: the case is badged `high`; it cannot approve, reject or pay anything	Provisional; changes ordering, not decisions
Legacy review threshold setting	0.30	`DUPLICATE_REVIEW_THRESHOLD`	Stored with the case and shown to the administrator as the `standard` band	Kept for compatibility with the settings screen

All three are readable and updatable at runtime through `GET` / `PUT /api/settings/threshold` (administrator only), and an update is written to the audit log. The values live in process memory (`backend/src/config/settings.js`), so a restart restores the environment defaults. That is a deliberate, documented limitation: a change made through the screen does not survive a redeploy.

6. Training data and provenance

Item	Value
Training file	`ml-service/training/realistic_pairs_train.csv` — 12 000 labelled pairs over 6 000 groups
Holdout file	`ml-service/evaluation/realistic_pairs_holdout.csv` — 12 000 pairs / 6 000 groups
Generator	`ml-service/generate_realistic_pairs.py`, seeded, with scripted corruption patterns
Positives	`repeat_application` only — one duplicate pattern
Negatives	Deliberately hard: siblings, twins and close ages, identifier collisions, common names, unrelated
Overlap between splits	0 shared groups, 0 shared identities, 0 identical rows (measured)
Holdout used for fitting	`false`, asserted by the training script
Holdout used for model selection	`true` — the same holdout chose XGBoost over Random Forest, so the reported numbers are selection-optimistic
Library versions	Python 3.11.9, XGBoost 3.2.0, scikit-learn 1.9.0, pandas 3.0.5, jellyfish 1.2.1, joblib 1.6.0
Seed	42

7. Measured performance (synthetic holdout — not an accuracy claim)

Per-pair results on the 12 000-pair holdout at each cut-off, from `ml-service/evaluation/operating_point_2026-10-30.json`:

Cut-off	Precision	Recall	F1	Miss rate	FPR on hard negatives	Review share
0.30	0.8694	0.9352	0.9011	6.48%	14.05%	53.78%
0.40	0.8985	0.9132	0.9058	8.68%	10.32%	50.82%
0.50 (live)	**0.9204**	**0.8927**	**0.9063**	**10.73%**	**7.72%**	**48.49%**
0.60	0.9404	0.8647	0.9009	13.53%	5.48%	45.97%
0.70	0.9607	0.8270	0.8888	17.30%	3.38%	43.04%
0.80	0.9791	0.7797	0.8681	22.03%	1.67%	39.82%
0.90	0.9899	0.7037	0.8226	29.63%	0.72%	35.54%

Ranking quality: ROC-AUC 0.9685, PR-AUC 0.9727. Best F1 is 0.9065 at 0.35; the live 0.50 scores 0.9063 and has the highest Youden J (0.8155). F1 is flat across 0.30-0.60, which is why the cut-off is described as provisional rather than tuned.

Overfitting checks (`docs/ml-training-integrity-2026-10-30.md`):

Check	Result
In-sample F1 versus holdout F1	0.9196 versus 0.9063 — a 1.33-point gap
5-fold grouped cross-validation inside the training file	0.9067 mean F1, within 0.0004 of the holdout

Check	Result
Learning curve (10% → 100% of groups)	0.8886 → 0.9063 F1, close to flat
Per-feature ROC-AUC	0.34-0.71; no feature approaches 0.95, so there is no generator shortcut

The independent per-pair false-positive baseline on 2 000 synthetic unrelated pairs (`docs/ml-evaluation-baseline-2026-10.md`) is **0.35% at cut-off 0.50** (7 pairs), with 0.15% when names are distinct. The 7.72% figure in the table above is measured on a negative set enriched with confusable cases and is **not** interchangeable with 0.35%; both must be reported together or neither.

8. From score to case: the registration decision path

Registration never compares only a candidate subset; it scores the new record against every existing student and then applies an aggregation policy.

Step	Implementation	Output
1. Lock	`pg_advisory_xact_lock(hashtext('sbms.student.registration'))` inside a transaction	Serialises concurrent registrations so two registrations cannot interleave their comparisons
2. Candidate set	`SELECT ... FROM students ORDER BY created_at ASC, id ASC FOR UPDATE`	Every existing record, frozen for the transaction
3. Scoring	`scorePairs([{first: submitted, second: candidate}])` in one Python call	A score and feature vector per candidate
4. Exact identifier evidence	`exactReviewIdentifierMatches` compares UPI, birth certificate and national ID	`exact_identifier_matches[]` per case
5. Qualification	Score >= `ML_REVIEW_SCORE_THRESHOLD` **or** an exact identifier agreement	Qualifying candidates
6. Aggregation	`summarizeAggregation` with policy `top-1` (default) or `all`	At most one case, or one case per pair
7. Persistence	`INSERT INTO duplicate_flags ... ON CONFLICT DO NOTHING`	One row per surviving candidate
8. Audit	`duplicate_candidates_considered`, `_qualifying`, `_suppressed`, `_flags_created`, model version, threshold, policy	Audit row plus the `duplicate_screening` response block

The `top-1` policy ranks exact identifier agreement above score, then score. Its purpose is to stop the review queue growing with registry size: with one case per registration the queue grows linearly in *registrations* rather than in *registrations* × *registry size*.

The arithmetic that motivated the policy, using the measured per-pair rate p = 0.0035 at cut-off 0.50:

Existing students N	Expected flags per registration before the fix	Probability of at least one nuisance flag
100	0.35	29.6%
500	1.75	82.7%
1 000	3.5	97.0%
5 000	17.5	~100%

Holding a 1% registration-level nuisance rate at N = 1 000 would need p <= about 0.001%, roughly 350 times lower than the measured 0.35%. Cut-off tuning therefore cannot substitute for better features, blocking or the corroboration policy. The `top-1` change caps the number of cases per registration; it does **not** reduce the probability that a particular legitimate registration receives one.

9. Review policy and what the administrator sees

Field	Computed by	Meaning
`risk_level`	fixed `review`	A case exists and no decision has been made. There is deliberately no `critical` level, because no score makes a decision
`priority`	`reviewPriority({score, highPriorityThreshold, exactIdentifierCount})` in `mlDuplicateReview.js`, shared by the registration response and the queue	`high` when an identifier agrees exactly or the score reaches 0.70; otherwise `standard`
`exact_identifier_matches`	`exactReviewIdentifierMatches`	`['UPI']`, `['birth certificate number']`, `['National ID']` and combinations
`review_reason`	registration controller	An exact-match sentence naming the identifier, or a sentence naming the provisional cut-off
Stored provenance	`duplicate_flags` columns	`model_version`, `review_threshold`, `comparison_features`
Masked identifiers	`duplicateFlags.controller.js:37-42`	Only the last three characters are shown until an audited reveal

The officer's confirmation screen shows plain language and no model internals: a possible-match sentence, the exact-identifier sentence when applicable, and counts ("checked against N existing students; M flagged; K recorded in the audit log without a separate case"). The administrator's queue shows the score, model version, both cut-offs and the aggregation policy, with the explicit statement that the thresholds are provisional.

10. Research artefacts deliberately excluded from the deployed claim

Artefact	Why it cannot be used as evidence for the deployed system
`ml-service/experiments/sbms_dataset/` (ten-feature schema, 16 800/2 400/4 800 split)	Adds four presence and conflict flags and drops `upi_match` and `birth_certificate_match`; the archive has no UPI or birth-certificate values and its school labels are not the five SBMS schools, so the artefact cannot be loaded by `predict.py`
Supplied-archive results (XGBoost precision 0.906, recall 0.846, PR-AUC 0.953 at 0.50)	Properties of a different, incompatible synthetic archive at 25% positive prevalence; the research model was never integrated
`realistic_pairs_xgboost_training_metrics.json` older generations	Superseded; the current holdout figures are the ones in §7, and the integrity checker fails the build if the artifact hash stops matching the metrics
`ML_RELIABILITY_REPORT.md` Random Forest figures	Describe a candidate that the application never loads

Reporting any of these as the production model's performance would be false. They are useful only as comparison evidence and are labelled as research in Chapter Five.

11. What this subsystem cannot support, and why

Claim	Status	Blocked by
"The system detects duplicates accurately"	**Cannot be supported**	No adjudicated labels exist (D5); every label is synthetic
Real-world precision, recall or miss rate	Unmeasured	As above
Applicant-population false-positive rate	Unmeasured	The negative sets are synthetic and either random or deliberately enriched
"The 0.50 cut-off is the right one"	Unsupported	F1 is flat from 0.30 to 0.60; no owner cost policy exists (D2/D3)
Duplicate patterns other than `repeat_application`	Untested	The generator produces one positive pattern
Generalisation beyond the simulator	Unknown	Synthetic name, school and identifier distributions

Claim	Status	Blocked by
Stability across retraining	Unknown	One artifact, one seed; any retrain invalidates the recorded numbers
Real-world latency at registry scale	Unmeasured	Screening is linear in N; the only timing recorded is 22.67 ms for 50 pairs

Two further limitations belong in the report even though they are not model weaknesses. First, registration compares against the entire registry, so the cost per registration grows with the register and the administrator-initiated rescan is quadratic (FR-34). Second, the review queue's ranking is by score and exact-identifier agreement only; it has no notion of reviewer workload, so a surge produces a backlog rather than a prioritised list.

12. Decisions still owned by the programme owner

ID	Decision	Why the model cannot close it
D1	Adjudication protocol and reviewers	Only adjudicated pairs can produce real precision and recall
D2	Acceptable miss rate	A trade-off between missed duplicates and review volume; a value judgement
D3	Review capacity	Determines which cut-off is affordable
D4	Candidate blocking	A behavioural change to the controller; requires approval and new tests
D5	Adjudication ownership and data-protection basis	Required before any real pair is labelled

Providing a recommended default for each of D1-D5 is legitimate; presenting a default as the owner's decision is not. `docs/ml-decision-sheet-2026-10-30.md` keeps that distinction explicit, and this file follows it.

13. Reproduction

```
cd ml-service
.\.venv\Scripts\python.exe -m unittest discover -s tests -v           # 33 tests
.\.venv\Scripts\python.exe .\check_training_integrity.py              # leakage and hash audit, ~30 s
.\.venv\Scripts\python.exe .\evaluate_operating_point.py              # writes the operating-point JSON, ~25
s
.\.venv\Scripts\python.exe .\evaluate_false_positive_rate.py --output .\evaluation\fpr_baseline_2026-10-
30.json
.\.venv\Scripts\python.exe -m unittest discover -s experiments/sbms_dataset -p test_experiment.py   # 7
research tests
```

All five commands are read-only with respect to the artifact: none trains, overwrites or serialises a model. The integrity checker exits non-zero if the committed report stops describing the live artifact, if the holdout and training splits overlap, if serving features drift from the generator's, or if a feature alone predicts the label.

14. Evidence index

Statement in this file	Source
Six feature definitions and missing-value rules	`ml-service/feature_engineering.py:11-116`
Artifact path, version string, protocol	`ml-service/predict.py:14,50-111`
503 on predictor failure	`backend/src/controllers/students.controller.js:74-96`; `docs/FINAL_TEST_REPORT.md`
Cut-off defaults and runtime mutation	`backend/src/config/settings.js`
Holdout metrics at each cut-off	`docs/ml-decision-sheet-2026-10-30.md` §1
False-positive baseline	`docs/ml-evaluation-baseline-2026-10.md` §2
Leakage and learning-curve checks	`docs/ml-training-integrity-2026-10-30.md` §1-§3

Statement in this file	Source
`top-1` aggregation and its limits	`docs/ml-evaluation-baseline-2026-10.md` §4
Review priority and masking	`backend/src/services/mlDuplicateReview.js`; `docs/frontend-verification-2026-10-30.md` §1
Research incompatibility	`ml-service/reports/sbms_dataset_experiment.md` §"Compatibility and limits"